

Content Delivery Network (CDN)

Alessio Sacco, Marco Rapelli

Politecnico di Torino



1859

Copyright notice

- This set of transparencies, hereinafter referred to as slides, is protected by copyright laws and provisions of International Treaties. The title and copyright regarding the slides (including, but not limited to, each and every image, photography, animation, video, audio, music and text) are property of the authors specified on page 1.
- The slides may be reproduced and used freely by research institutes, schools and Universities for non-profit institutional purposes. In such cases, no authorization is requested.
- Any total or partial use or reproduction (including, but not limited to, reproduction on magnetic media, computer networks, and printed reproduction) is forbidden, unless explicitly authorized by the authors by means of written license.
- Information included in these slides is deemed as accurate at the date of publication. Such information is supplied for merely educational purposes and may not be used in designing systems, products, networks, etc. In any case, these slides are subject to changes without any previous notice. The authors do not assume any responsibility for the contents of these slides (including, but not limited to, accuracy, completeness, enforceability, updated-ness of information hereinafter provided).
- In any case, accordance with information hereinafter included must not be declared.
- In any case, this copyright notice must never be removed and must be reported even in partial uses.

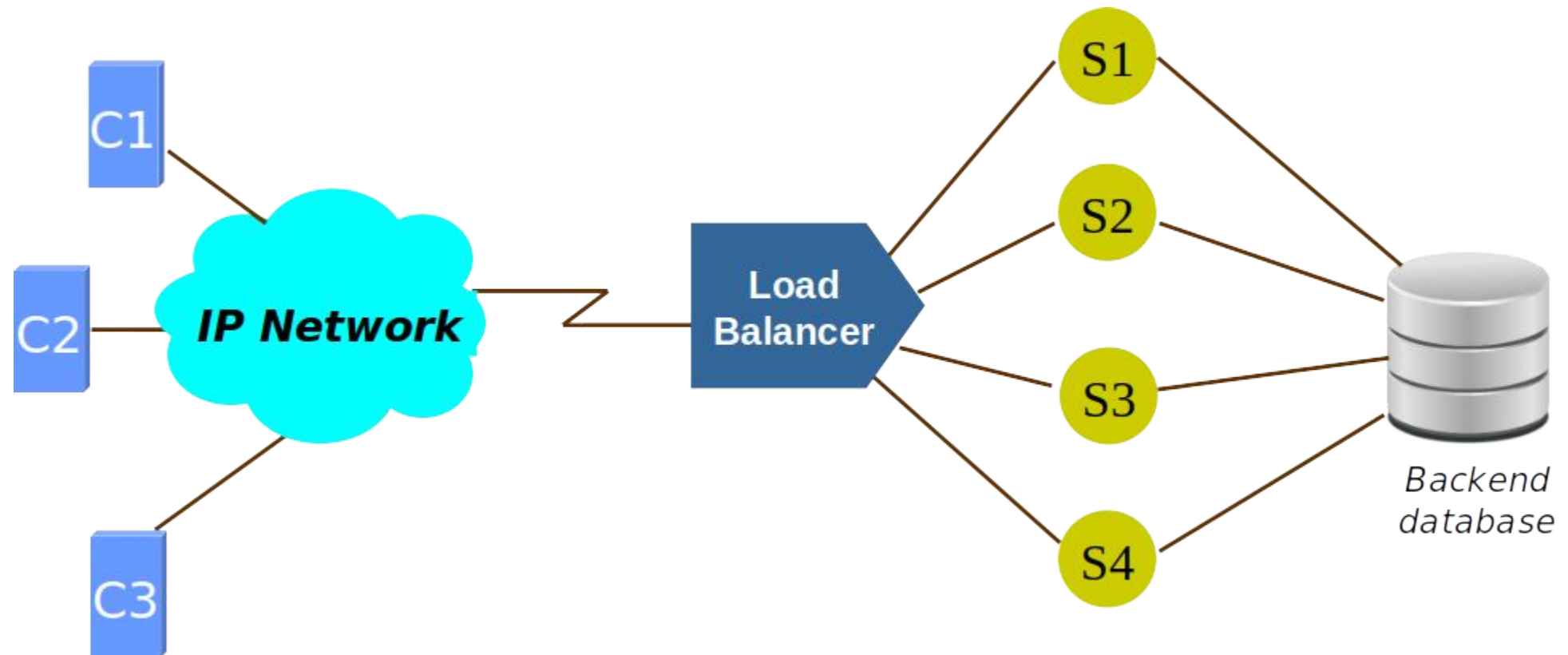


Introduction

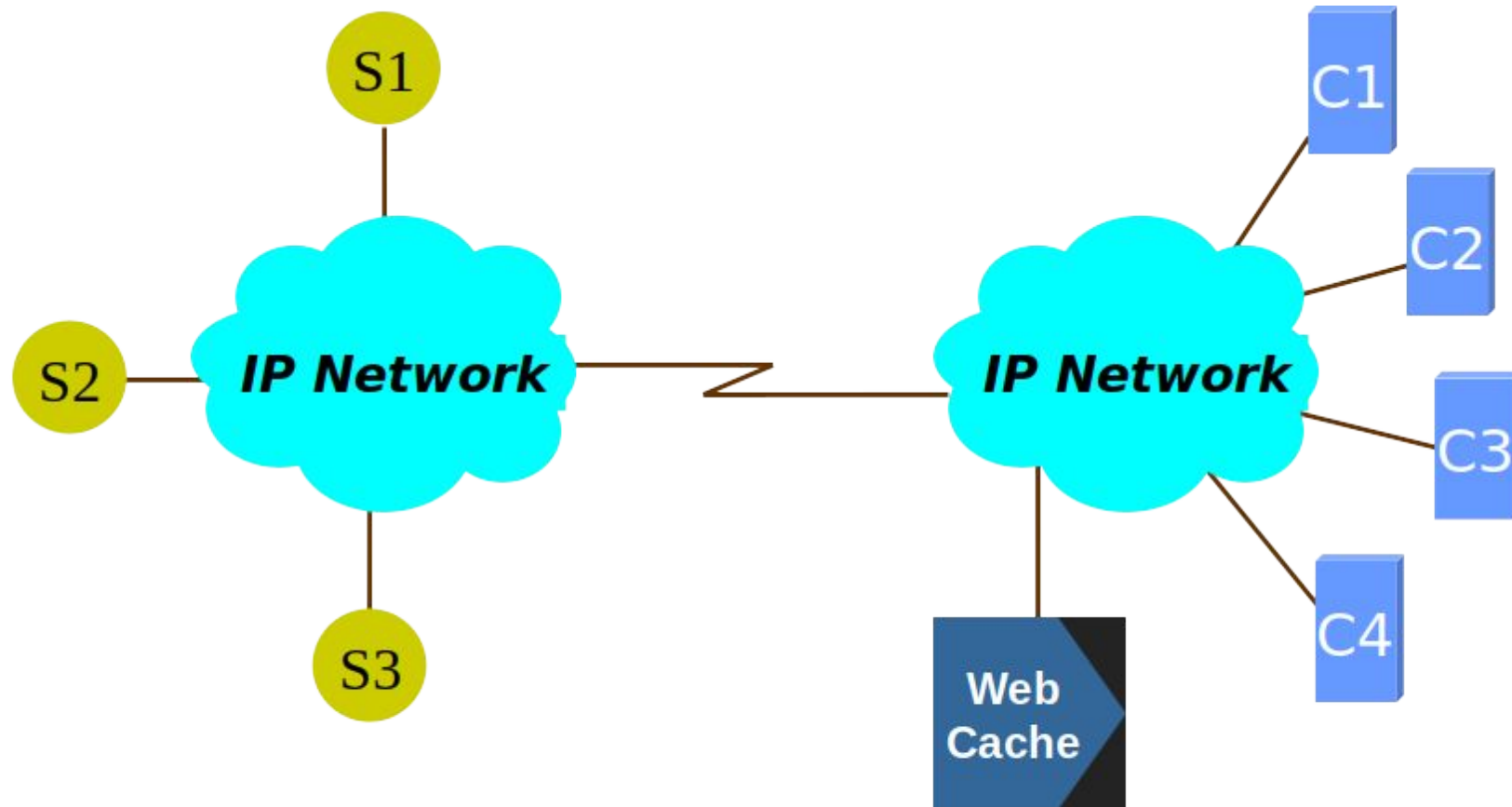


How to deploy large-scale web servers

- Deploy and design efficient server farms:
 - Front-end load-balancers requests over servers
 - Servers access the same backend database



Old days: web Caches



A Web Cache is a device that stores a local copy of more recently required HTTP objects and reacts as proxy server to clients' requests

Why caches?

- (1) Keep traffic local, decreasing the pressure (i.e., cost) on Internet transit links
 - Connection points between ISP and Internet backbones may be more overloaded than intra-ISP links
 - Cost of transferring traffic over Internet backbones is usually more expensive than intra-ISP links (the former may also depend on the volume of traffic, while the latter is usually “flat”)
- (2) Better Quality of Experience for end users
 - Decrease latency, hence more prompt answer from remote web server (e.g., limits the impact of the three-way handshake)
 - Improves throughput (TCP throughput is proportional to WND/RTT)
- (3) Enables localized content (e.g., advertisement of local shops)



HTTP Caching

- Clients often cache documents
 - When should origin be checked for changes?
 - Every time? Every session? Date?
- HTTP includes caching information in headers
 - HTTP/1.0 used: “Expires: <date>”; “Pragma:no-cache”
 - HTTP/1.1 has “Cache-Control”
 - “No-Cache”, “Private”, “Max-age: <seconds>”
 - “E-tag: <opaque value>”
- If not expired, use cached only
- If expired, use condition GET request to origin
 - “If-Modified-Since: <date>”, “If-None-Match: <etag> ”
 - 304 (“Not Modified”) or 200 (“OK”) response



Historical perspective

- Web proxies (1990)
 - Need to be explicitly configured from end-users
 - The usual problem of asking user to do something (they may forget about this, they may input different parameters, etc)
- Transparent proxies
 - Transparent TCP/HTTP proxies are widely used nowadays
- SSL breaks the previous model
 - Selling point: your connection is more secure, because nobody can inspect what you are doing (only the final endpoint, e.g., Google, can)
 - Actual point: nobody in the path can understand what you are doing, hence ISP become blind
 - Necessity of explicit agreements between Over-The-Top (i.e., content providers) and ISP to setup their caches in the telco network backbone



Problems with these approaches

- Server farms do nothing about network congestion, or to improve latency issues due to the network
- Caching proxies serve only their clients, not all users on the Internet
- Content providers cannot rely on existence and correct implementation of caching proxies
- Accounting issues with caching proxies. For instance, www.cnn.com needs to know the number of hits to the webpage for advertisement displayed on the webpage
- Significant fraction (>50%) of HTTP objects are un-cacheable

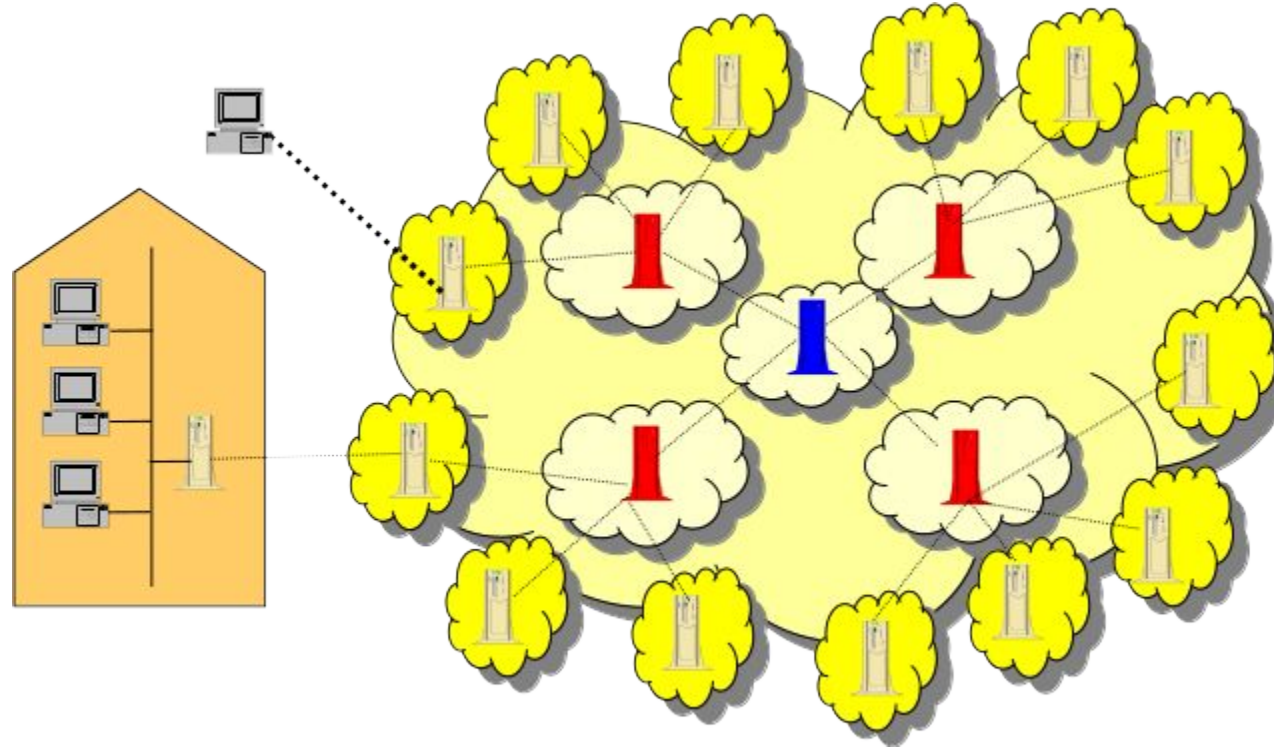


Content Delivery Networks



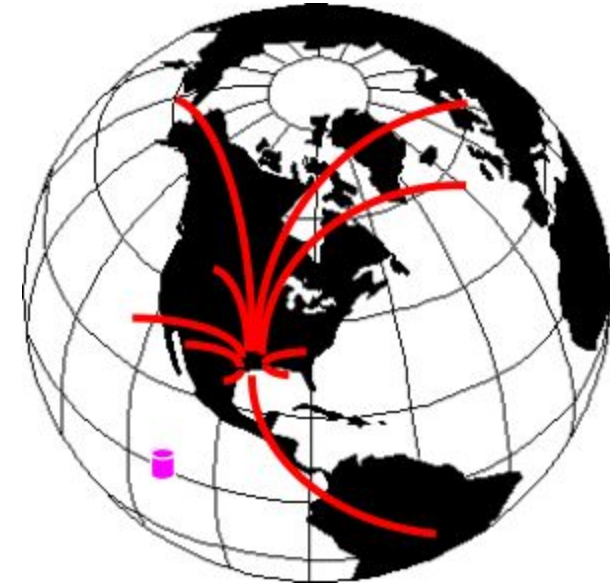
CDNs: An Overlay Network over Internet

- A Content Delivery Network (CDN) is an overlaid network of **Caches**, a.k.a. **Content Servers**, a.k.a. **Delivery Nodes**, a.k.a. **Replicas**



Motivations for CDNs

- An alternative way of adding caches in the Internet
- Web site performance improvement
 - Static Web pages or non-live streaming media
 - Server farms are far from users
 - Better RTT → better user's experience
- Bandwidth
 - Cost saving
 - Less probability to incur in a bottleneck
- Availability
- Load balancing
 - Traffic peaks crashes sites

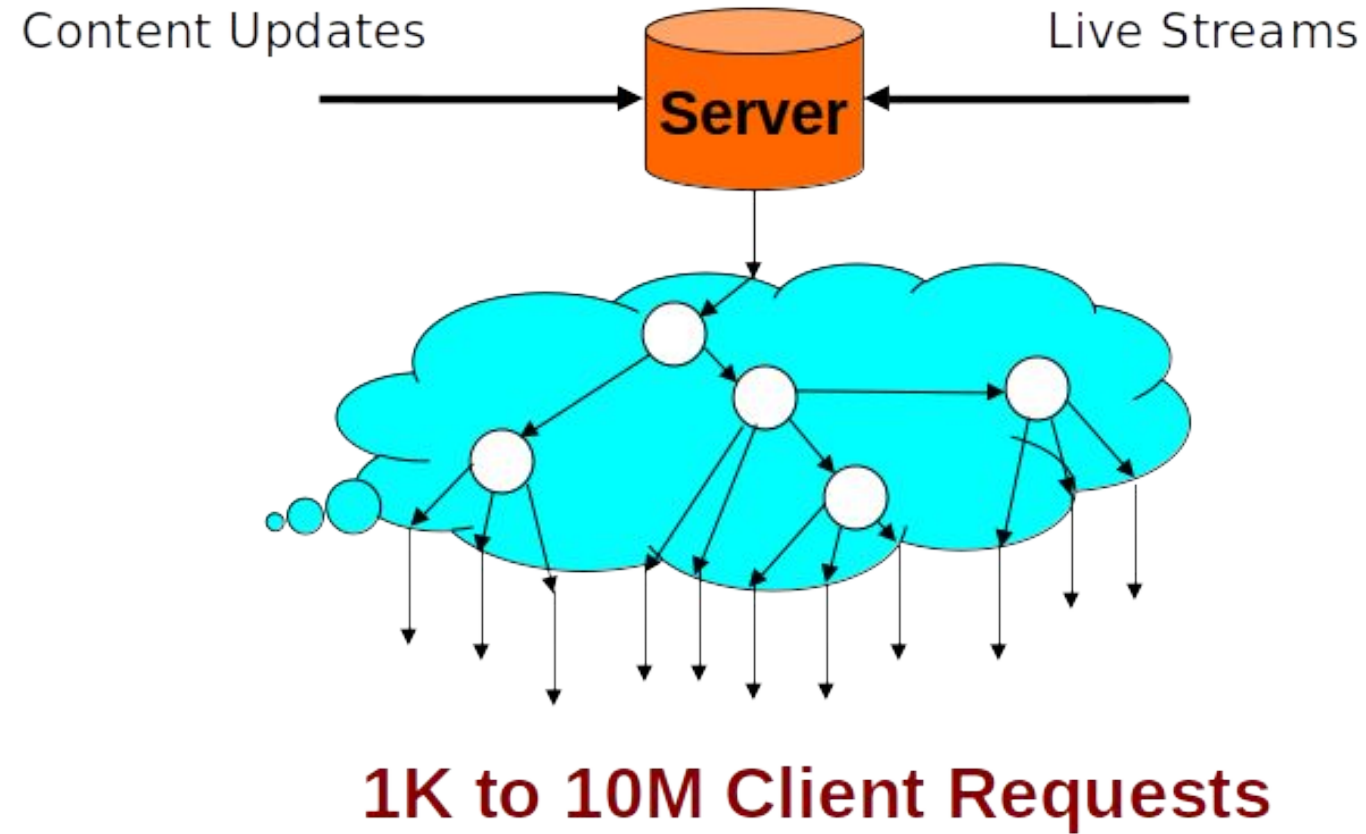


Usage Examples for CDNs

- Distributed Web Hosting
- Video-On-Demand
- Scalable Live Streaming
- Dynamic Content
- Conditional-Access Content
 - Advertisements



Example of a CDN used for Live Streaming



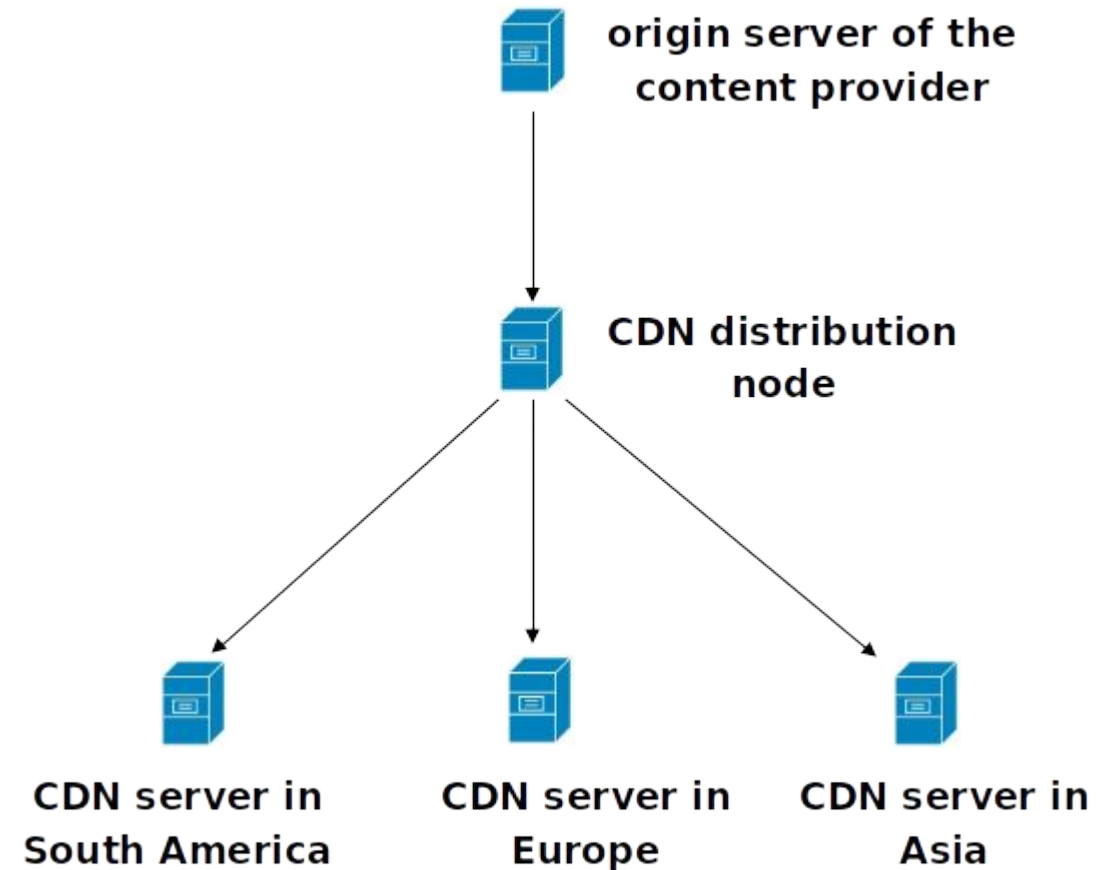
CDN vs Caching Proxies

- Caches are used by ISPs to reduce bandwidth consumption, CDNs are used by content providers to improve quality of service to end users
- Caches are reactive, CDNs are proactive
- Caching proxies cater to their users and not to content providers, CDNs cater to the content providers and clients
- CDNs give control over the content to content providers, caching proxies do not



Content Replication in CDNs (1)

- The CDN company installs thousands of servers throughout Internet:
 - In large data centers, or
 - close to users
- The CDN replicates customers' content
 - CDN customers and content providers!
- When the content provider updates the content, the CDN updates the servers.



Content Replication in CDNs (2)

- The origin server rewrites pages to serve content via CDN:

**Traditional web page
on a web server**

```
<html>
<head> <title> Fluffy Video </title> </head>
<body>
<h1> Fluffy Video's Product List </h1>
<p> Click below for free samples. </p>

<a href="koalas.mpg"> Koalas Today </a> <br>
<a href="kangaroos.mpg"> Funny Kangaroos </a> <br>
<a href="wombats.mpg"> Nice Wombats </a> <br>
</body>
</html>
```

**Web page that distributes
content via CDN**

```
<html>
<head> <title> Fluffy Video </title> </head>
<body>
<h1> Fluffy Video's Product List </h1>
<p> Click below for free samples. </p>

<a href="http://www.cdn.com/fluffyvideo/koalas.mpg"> Koalas Today </a> <br>
<a href="http://www.cdn.com/fluffyvideo/kangaroos.mpg"> Funny Kangaroos </a> <br>
<a href="http://www.cdn.com/fluffyvideo/wombats.mpg"> Nice Wombats </a> <br>
</body>
</html>
```



Content Replication in CDNs (3)

- The core idea of CDN consists in replicating content on many servers.
- However, the practical accomplishment of this idea encompasses multiple challenges:
 - How to replicate content
 - Where to replicate content
 - How to find replicated content
 - How to choose among server replicas
 - How to direct clients towards server replicas



Server Selection (1)

- Which server?
 - Lowest load: to balance load on servers
 - Best performance: to improve client performance
 - Any alive node: to provide fault tolerance
 - Based on Geography? RTT? Throughput? Load?
- How to direct clients to a particular server?
 - As part of routing: anycast, cluster load balancing
 - As part of application: HTTP redirect
 - As part of naming: DNS

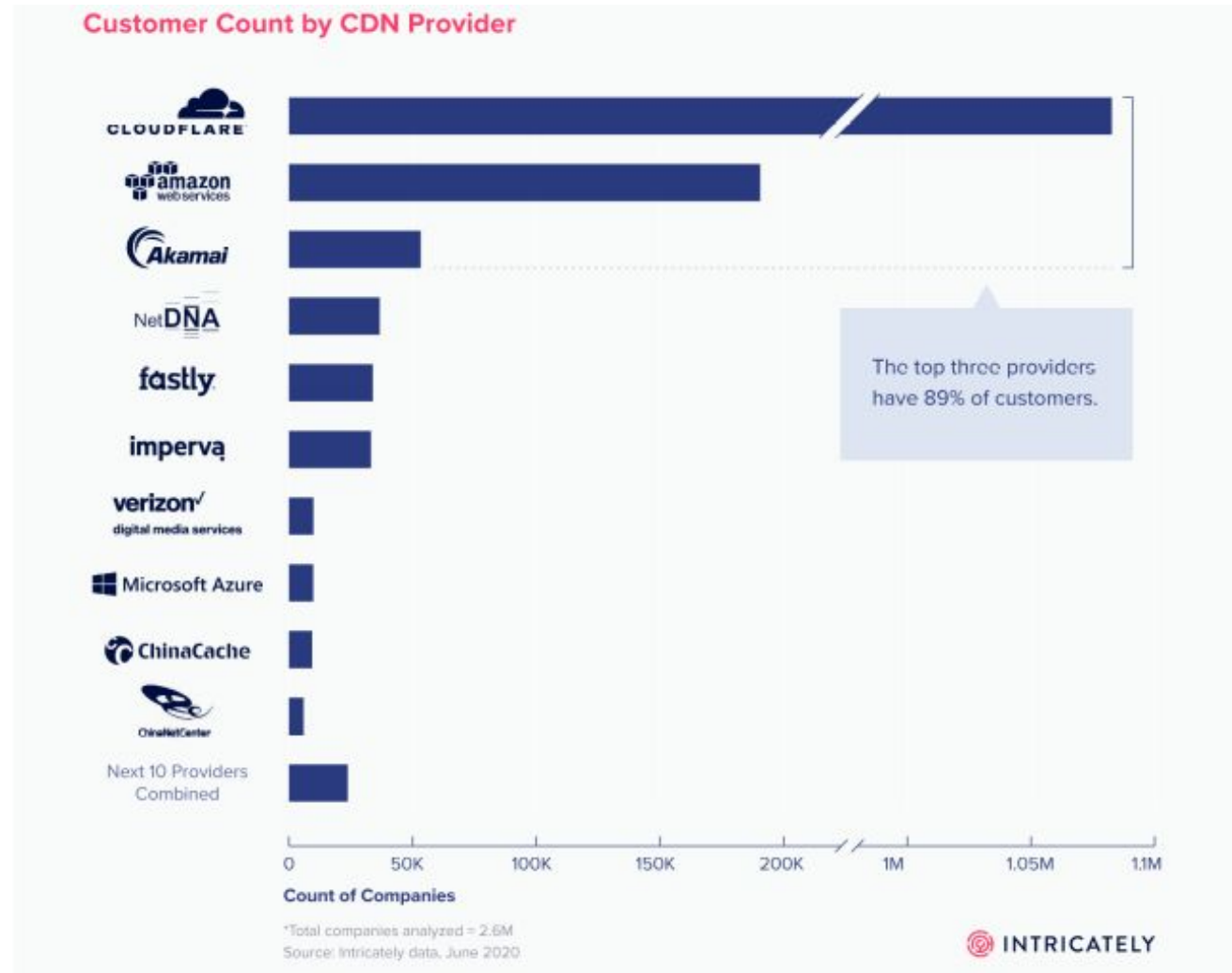


Server Selection (2)

- **Routing based** (IP anycast)
 - Pros: Transparent to clients, circumvents many routing issues
 - Cons: Little control, complex, scalability
- **Application based** (HTTP redirects)
 - Pros: Application-level, fine-grained control
 - Cons: Additional load and RTTs, hard to cache
- **Naming based** (DNS selection)
 - Pros: Well-suitable for caching, reduce RTTs
 - Cons: Request by resolver not client, request for domain not URL, hidden load factor of resolver's population



CDN market share



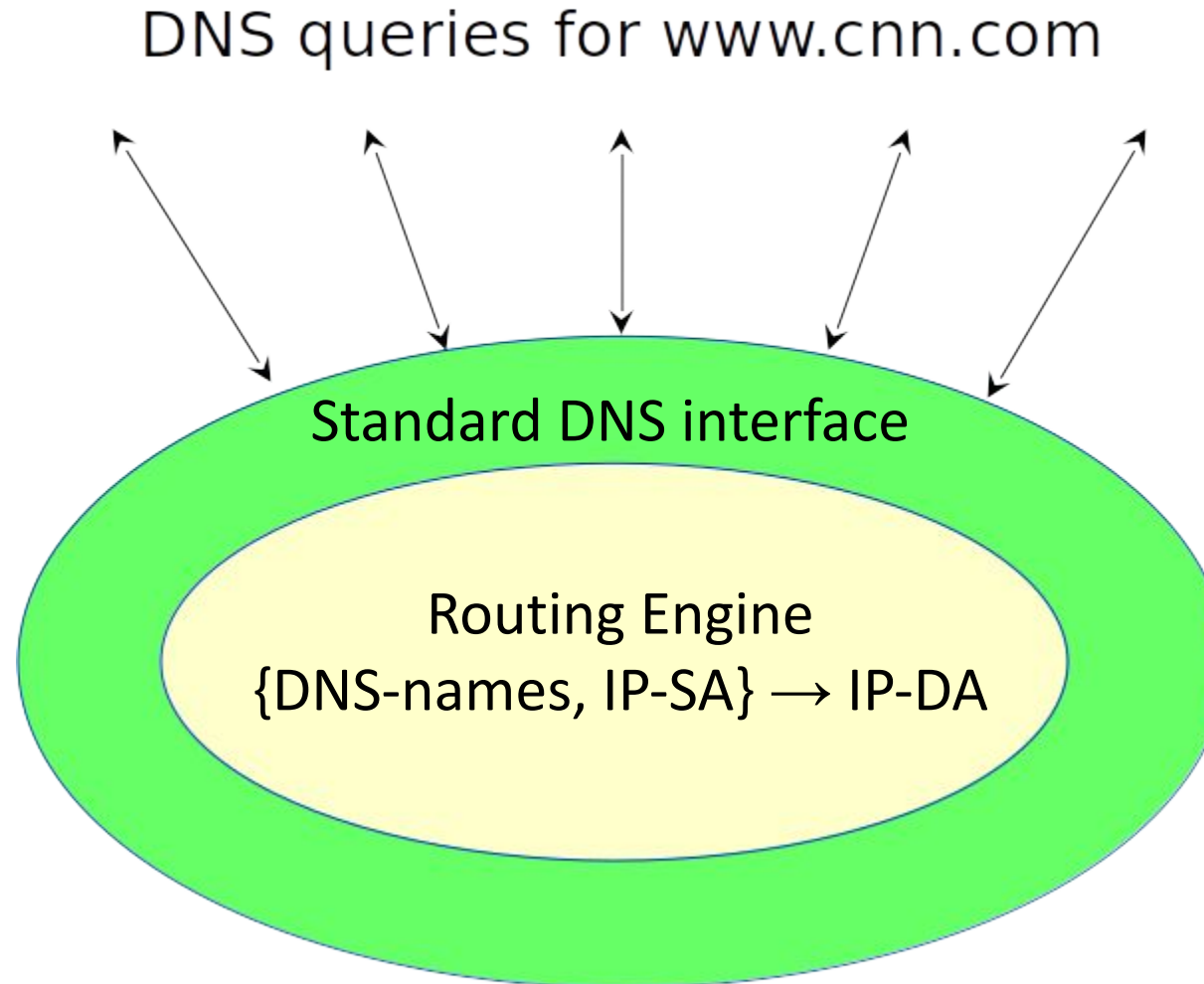
<https://blog.intricate.com/2020-state-of-the-cdn-industry-trends-market-share-customer-size>



DNS-based CDN



The idea: a new DNS Server Architecture

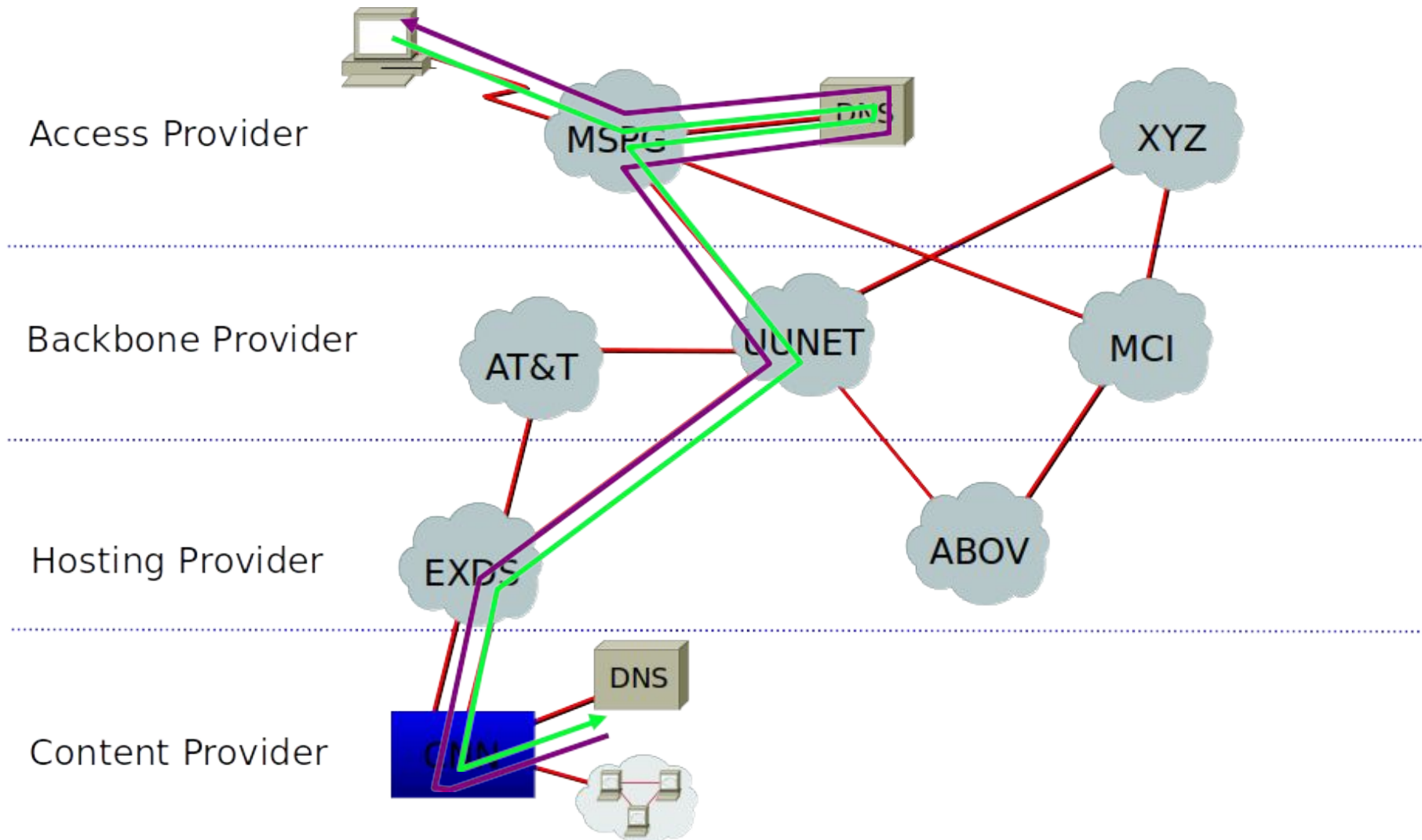


DNS-based CDNs

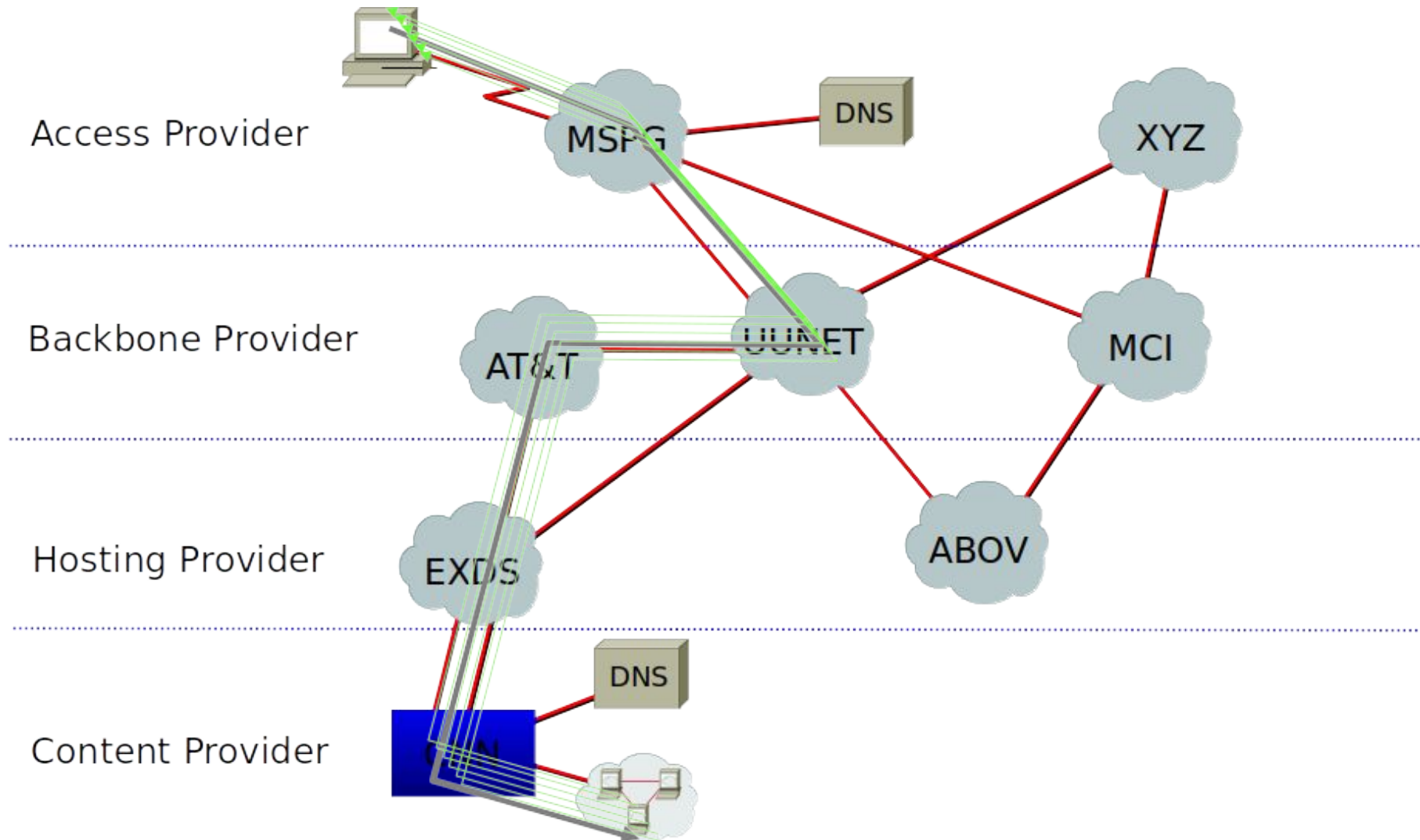
- Host Names are used to redirect the traffic to the best replica
 - the replica selection happens when the name is translated to an IP address
- DNS servers become “Content Routers”
 - they measure as many metrics as possible (RTT, Server Load, Layer 3 metrics, response time, etc.) to compute a **replica routing table** $\{\{\text{DNS-names, IP-SA}\} \rightarrow \text{IP-DA}\}$
 - Metric measurement is not easy
 - Layer 3 metrics alone are not particularly meaningful



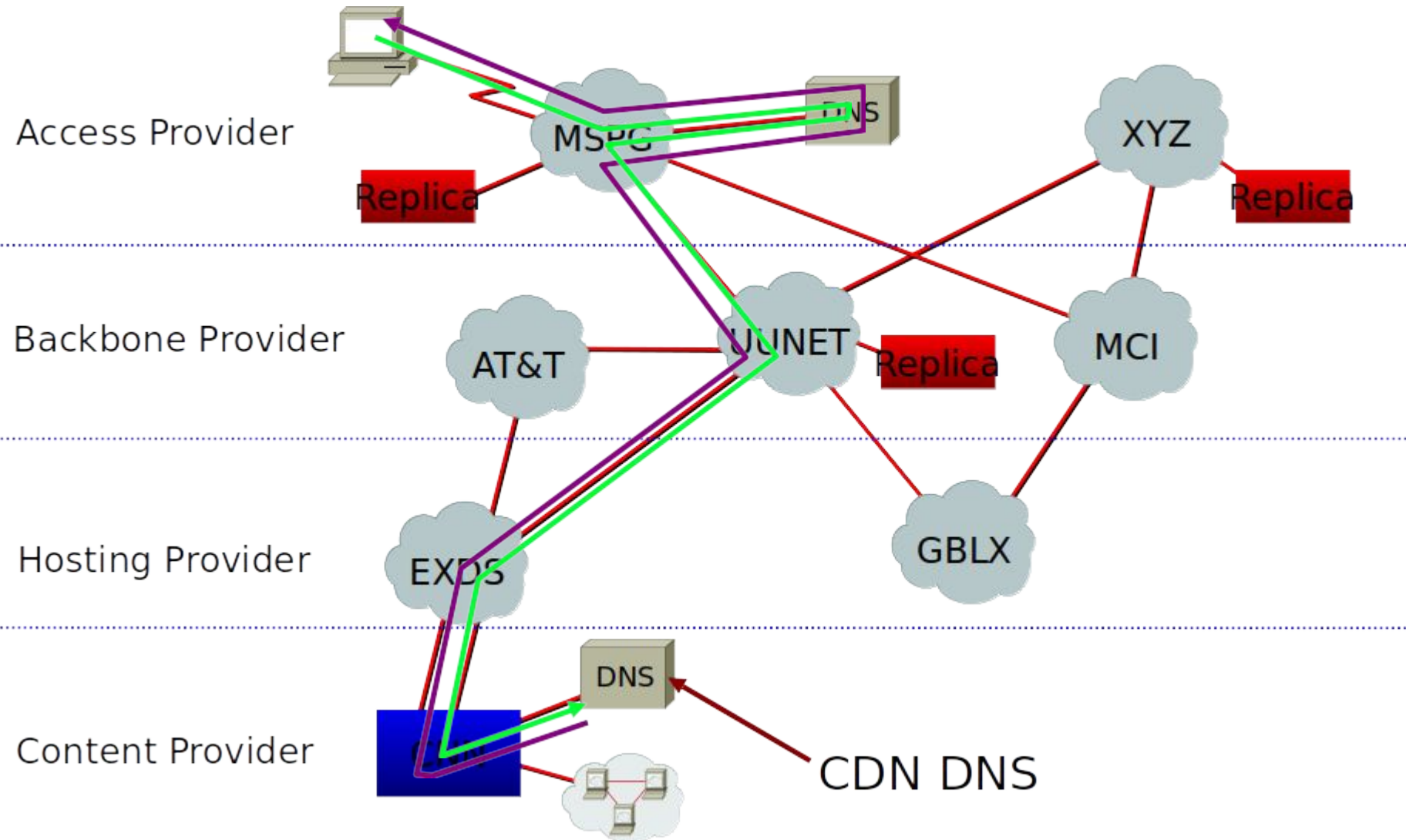
Traditional Browsing



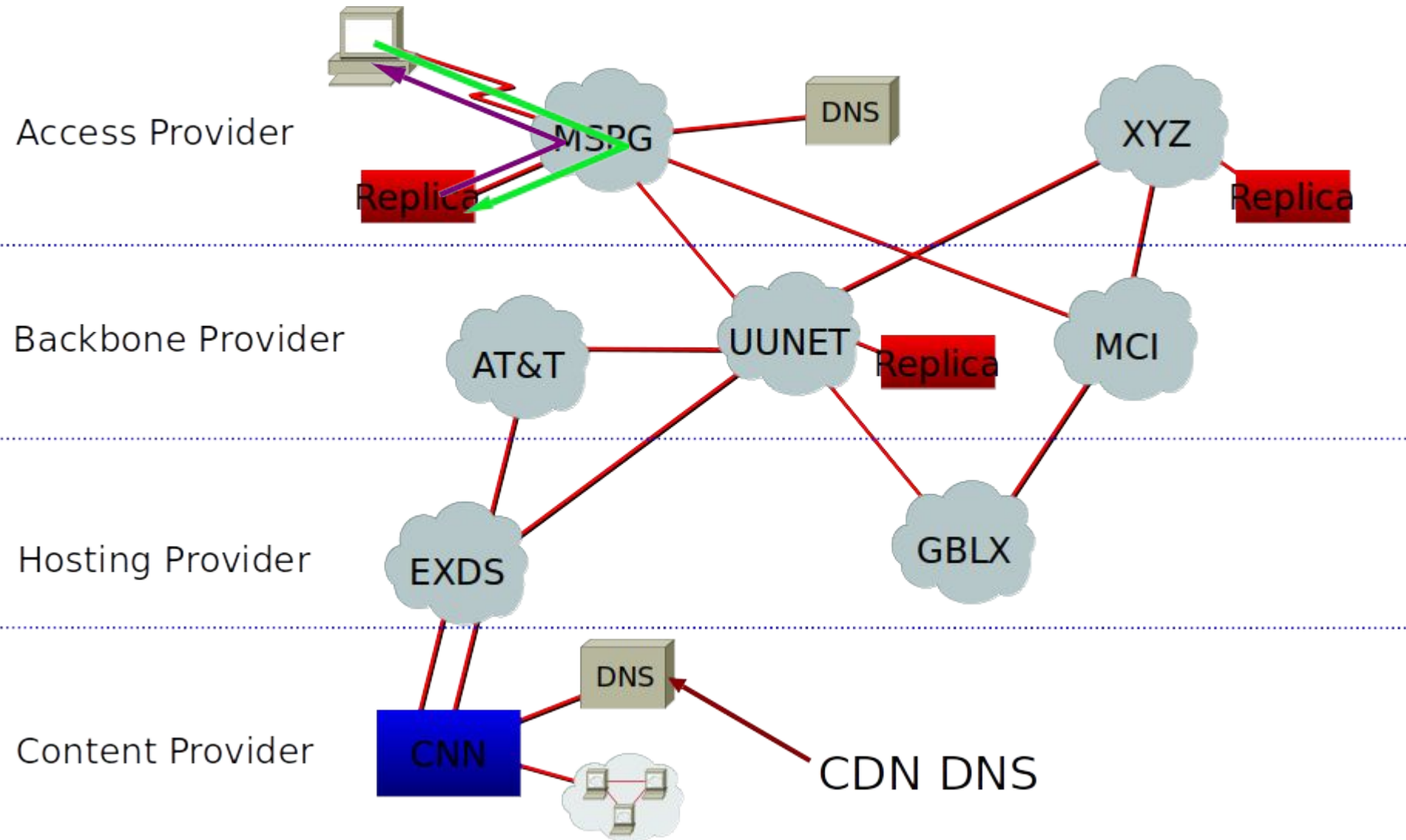
Traditional Browsing



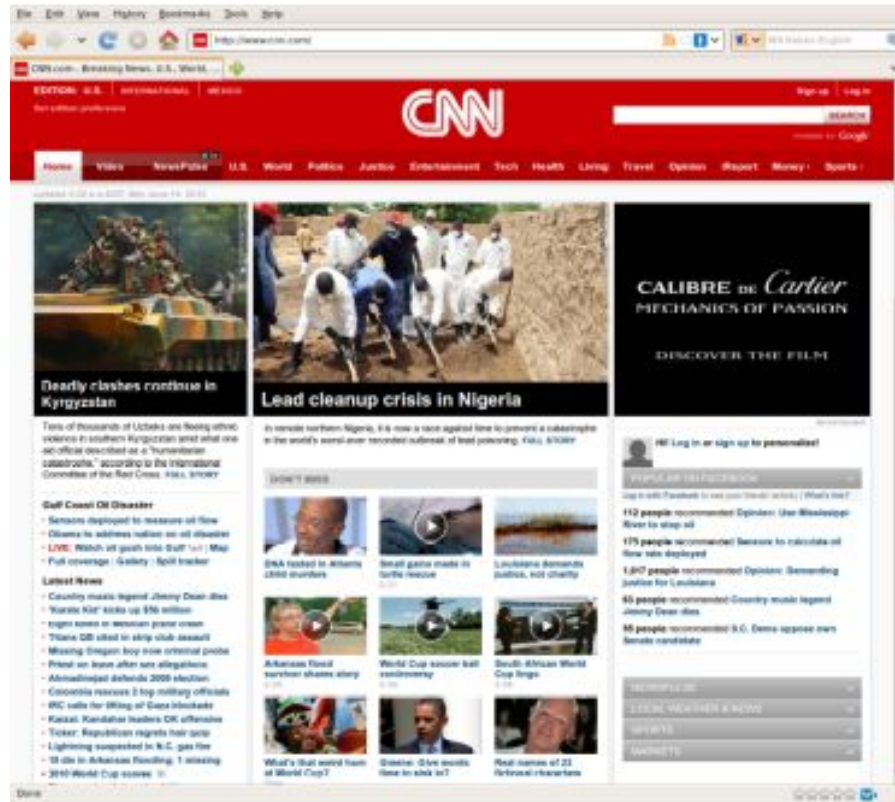
DNS-based CDN Browsing



DNS-based CDN Browsing



Example



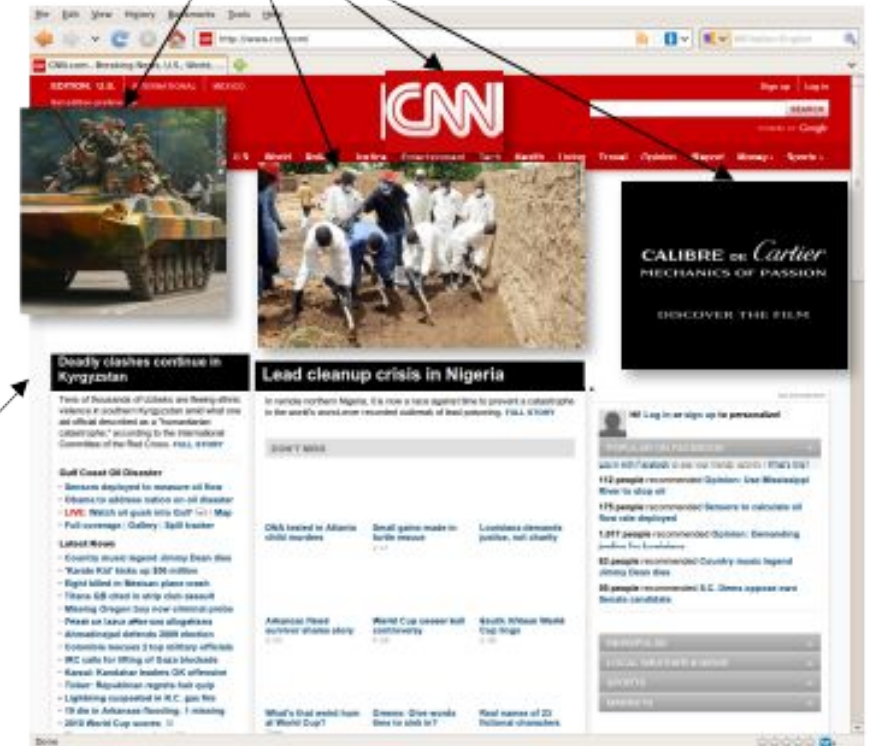
Entire page delivered by CNN

Akamaization



HTML delivered by CNN

"Akamaized" content



DNS-based CDNs Limitations

- There are limitations
 - The granularity of redirection is an host name, not a URL
 - Content of large web sites cannot be split into multiple caches
 - It is difficult to use the same host name for static and dynamic content
- The Akamai approach:
 - Akamaized URLs:
 - <http://a836.g.akamaitech.net/7/836/123/e358f5db0045e/www.cnn.com/logo.gif>



How Akamai works (1)

- It creates new domain names for each client content provider
 - E.g., a128.g.akamai.net
 - The CDNs DNS servers are authoritative for the new domains CDN's
- The content provider modifies its content so that embedded URLs reference the new domains
 - `http://cnn.com/a.gif` → `http://a128.g.akamai.net/7/23/cnn.com/a.gif`
 - Apache module or IIS ISAPI filter

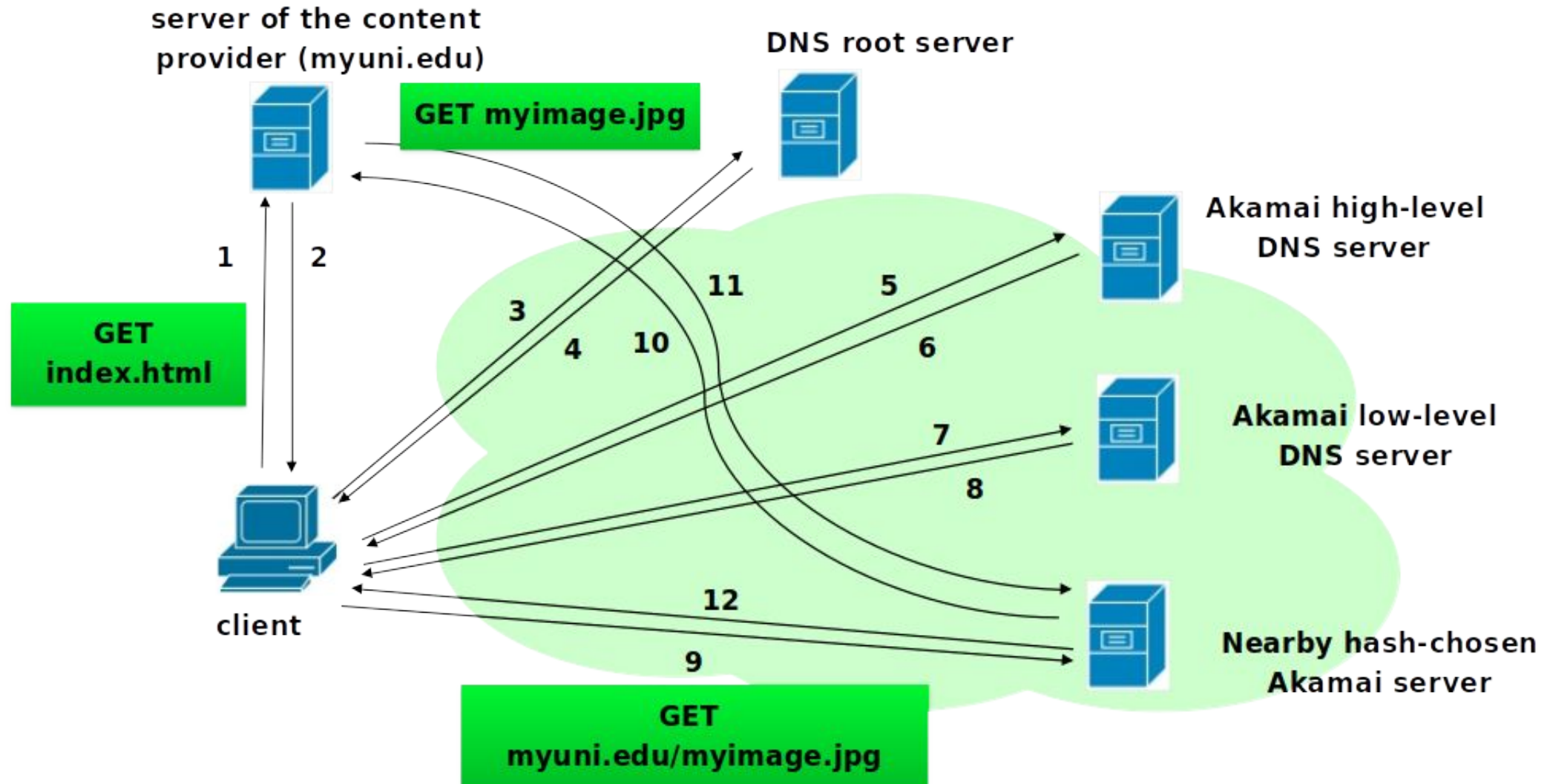


How Akamai works (2)

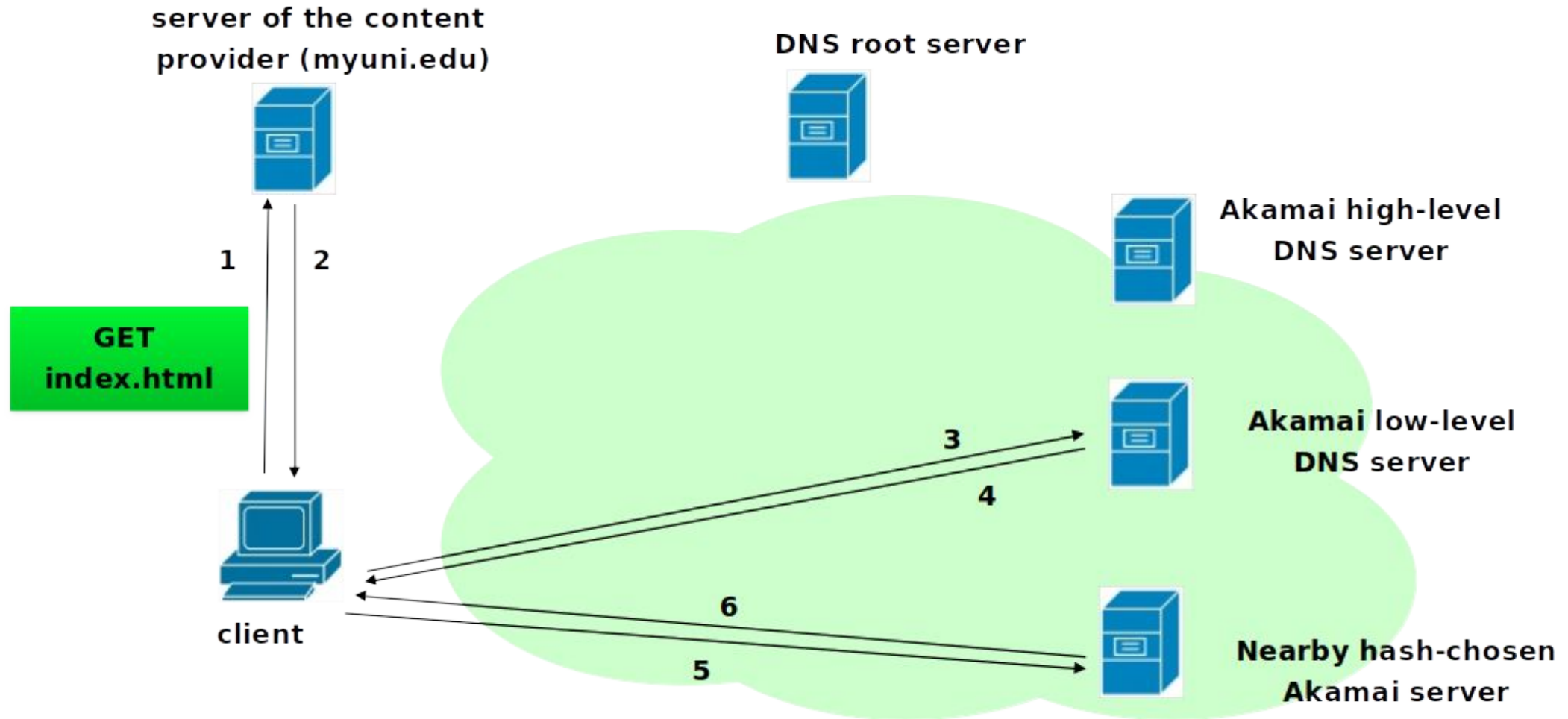
- Using multiple domain names for each client allows the CDN to further subdivide the content into groups
- DNS sees only the requested domain name, but it can route requests for different domains independently
 - `www.repubblica.it` may point to mainly Italian caches
 - `www.elpais.es` may point to mainly Spanish caches



Interaction with the Akamai CDN – Content Not Cached



Interaction with the Akamai CDN – Content Already Cached



Extension to DNS-based CDNs

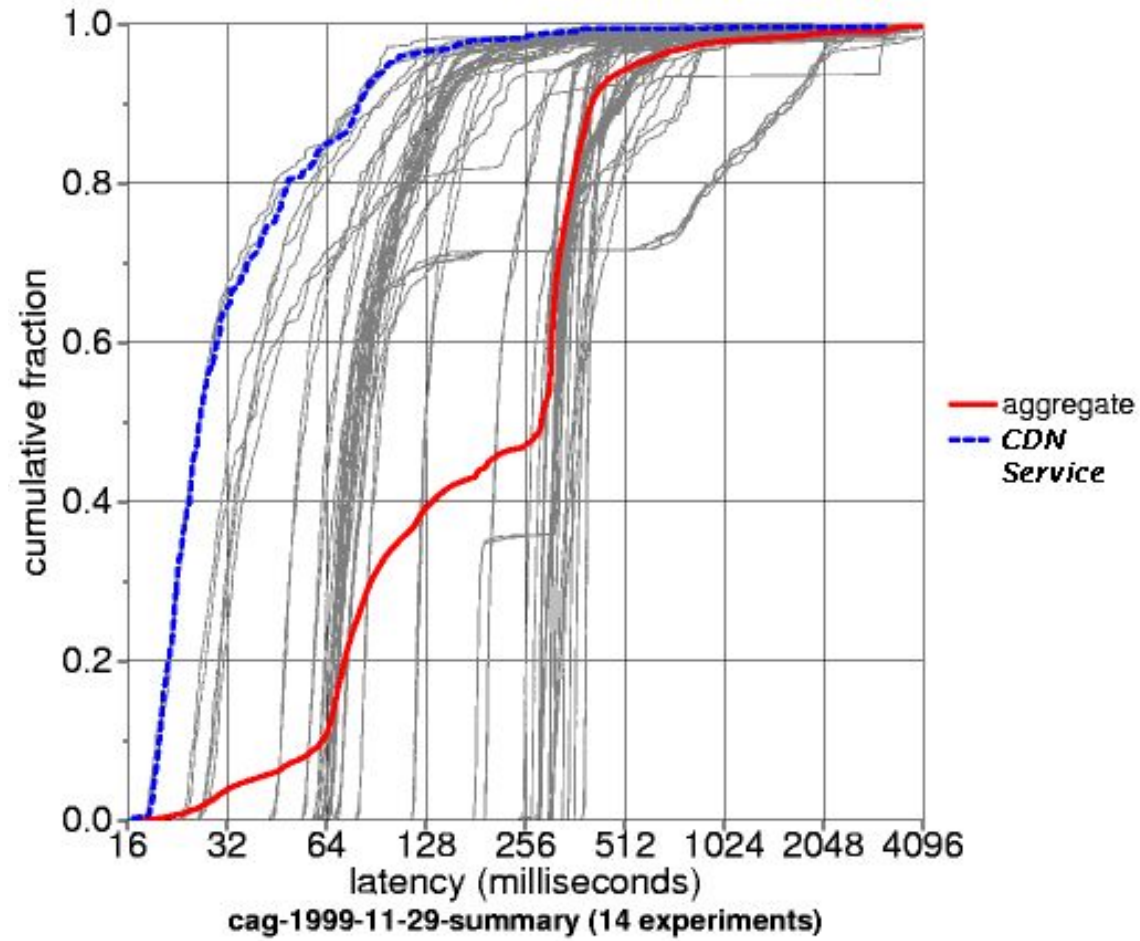
- How to implement more granular DNS-based CDNs (e.g. how to look for the complete URL)?
 - A possible answer: HTTP Redirect
- Redirection can be obtained in two ways
 - Every server in the farm is capable to redirect
 - An SLB (Server Load Balancer) is capable to redirect
- Requires always two requests per object
 - First toward the original site
 - Second toward the actual object URL
- Increased latency, hence possible effects on quality of experience from the user point of view
 - Limited effect if both objects are very close to the user, e.g., on the same enterprise network



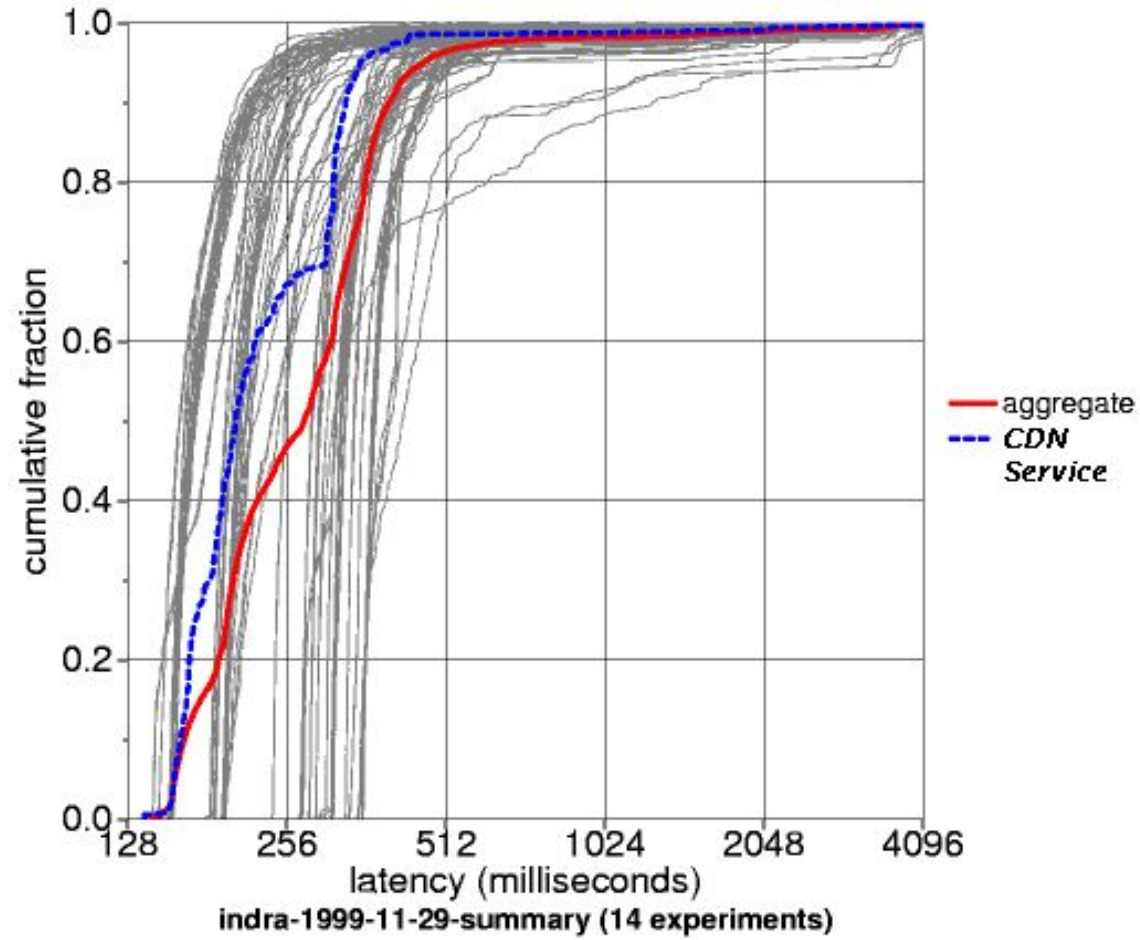
The Next Step: URL-based CDNs

- URLs are used to redirect the traffic to the best Content Server
- URL routing requires TCP termination
 - TCP termination is complex and expensive
 - TCP termination introduces delay
- There will be only one TCP termination point
 - Close to the client?
 - Close to the server?

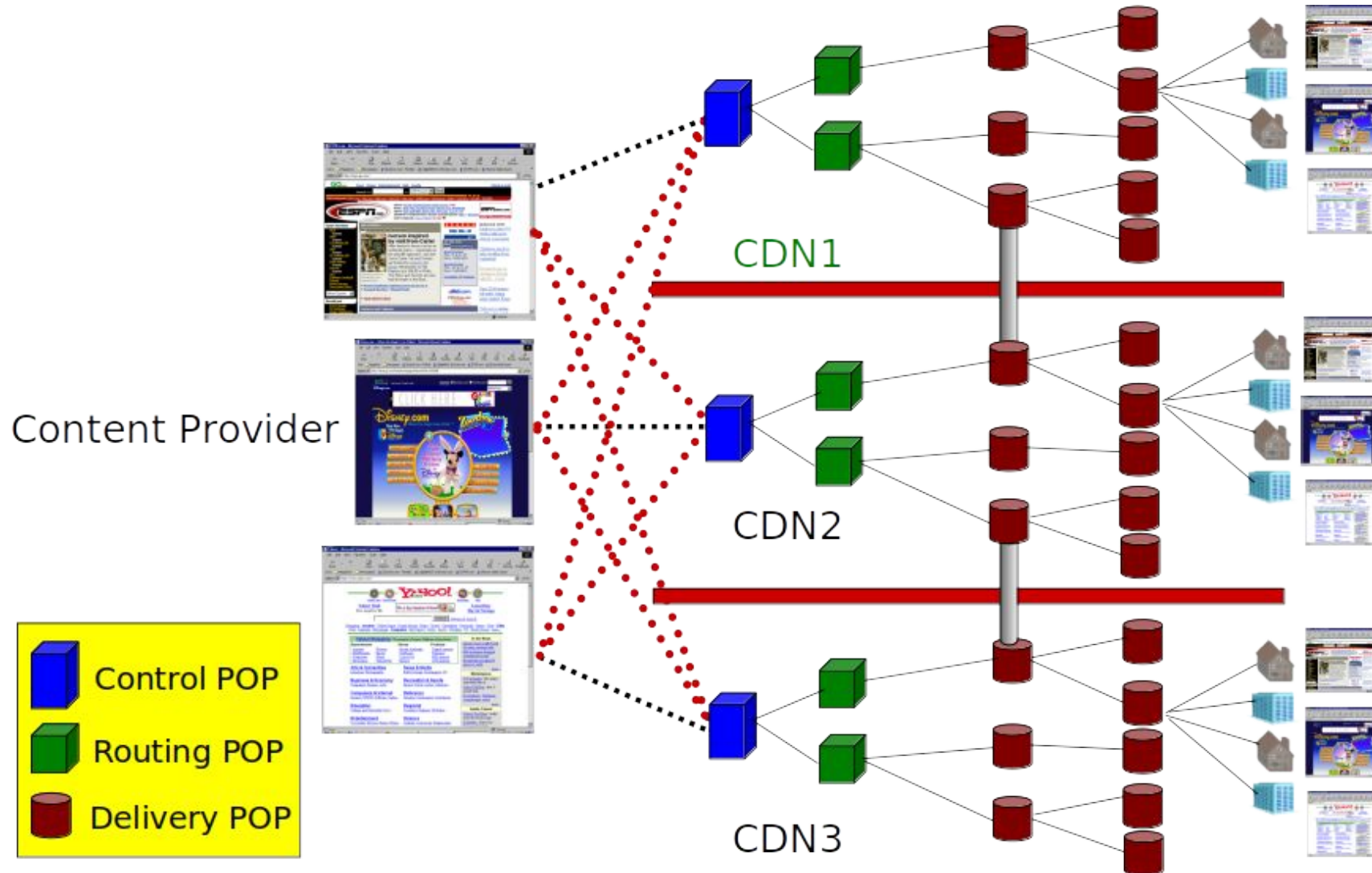
Sometime CDNs are very good!



Sometime are not so good!



Content Peering



Conclusions

- Content Delivery Networks (CDNs) DNS-based are widely deployed
 - CDNs are not only for web traffic, but also for multimedia streaming
 - Replicas will have slightly different content (e.g. local advertisement)
 - Content Peering is still an unsolved problem
- Some names
 - Akamai
 - Amazon CloudFront
 - Cloudflare
 - Limelight Networks

